# DMD User Guide


Installation 3
- Install App from the AppExchange 3
- Grant Permissions for App Setup 5
- Verify installation 6

Setup 7
- Allow DMD to securely store its Settings 7
- Authenticate with External APIs 8
- Adding  analysis component to record page 9
- Extend DMD app with choosen AI provider 10

Use the App 13
- Permissions layers 13
- Limitations 13
  - Document analysis request size 13
  - Document file type 13
- Known issues 13
- Prompts Recommendations 14

User Guide 15
- Creating Rules and Rulesets 15
  - Creating a Ruleset 15
  - Creating a Rule 16
  - Adding a Rule to A Ruleset 18
  - Removing a Rule from a Ruleset 19
  - Using Topic Tags to categorize Rules and Rulesets 19
- Check Documents by running an Analysis 20
- Using Regression Test suite 24
  - Configure regression test 24
  - Cancel regression test execution 26
- Export and Import Rulesets 26

The goal of this document is to provide information to administrators so that they can configure the DMD app in Salesforce.

The guide also provides instructions for end-users to help them get the most out of the functionality in the application.

DMD checks documents using Open AI GPT-4 model.
The document types and rules that you can run depend on your customization. See the user guide for details on how to customize the rules and rules sets that will be run against the documents.

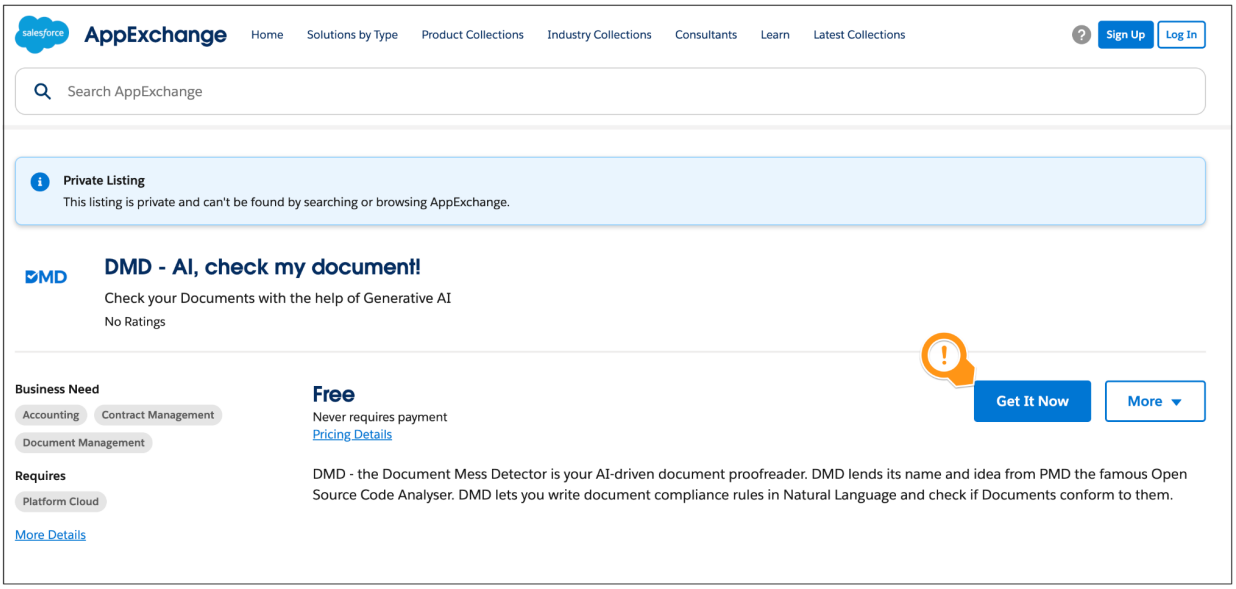
# Installation

Follow the steps below in the given order to install the DMD Salesforce app to your Salesforce environment.

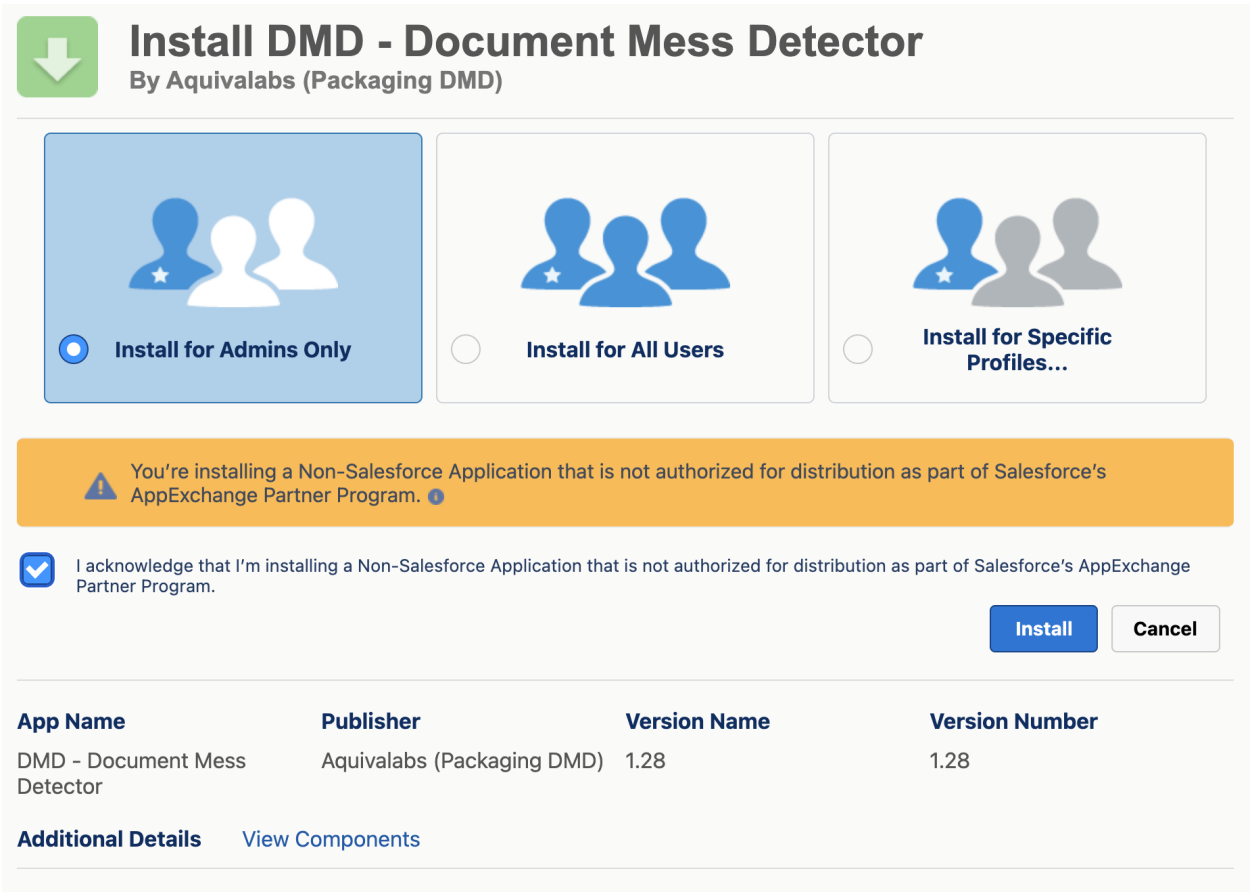
Most of those steps require you to be a System Administrator of your Salesforce production org.

## Install App from the AppExchange

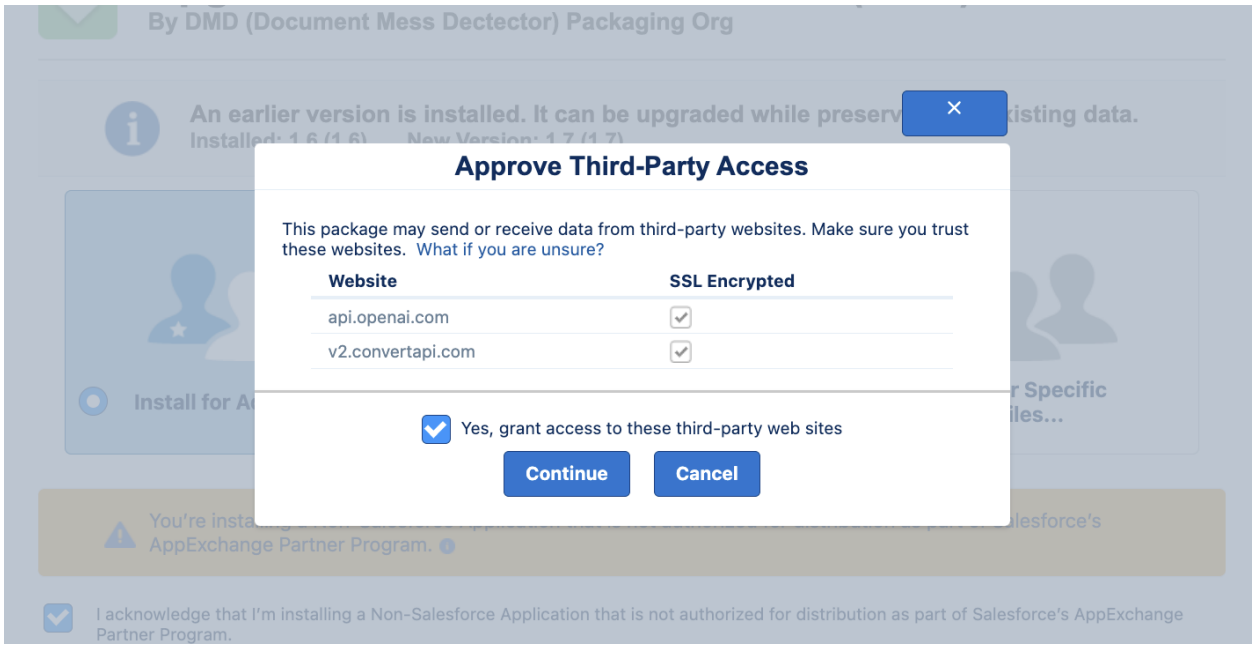
Load the AppExchange Listing page in your browser. You should see something like this. Click the big Install button.
It will ask you to log in to your Salesforce organization. Do that when asked for and continue the installation.



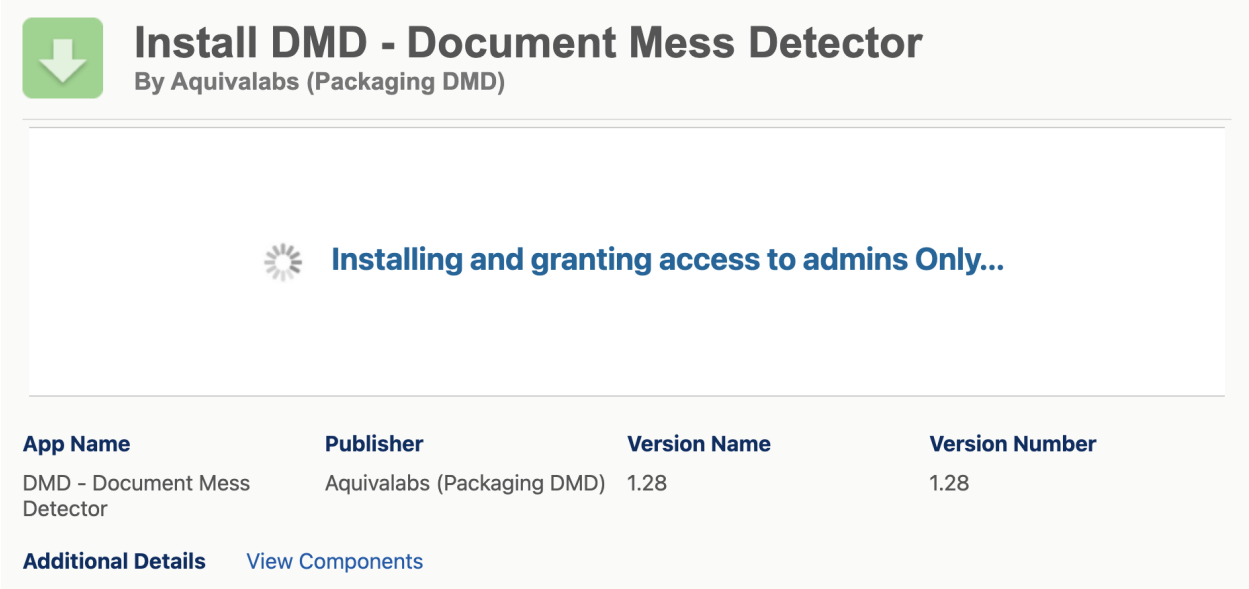

In the next screen, you will be asked who to grant access to the app. We recommend sticking with the default “Install for Admins Only.”




The App is making use of multiple external APIs and you need to approve the usage on this screen.



Once the installation has started, the following status will appear:

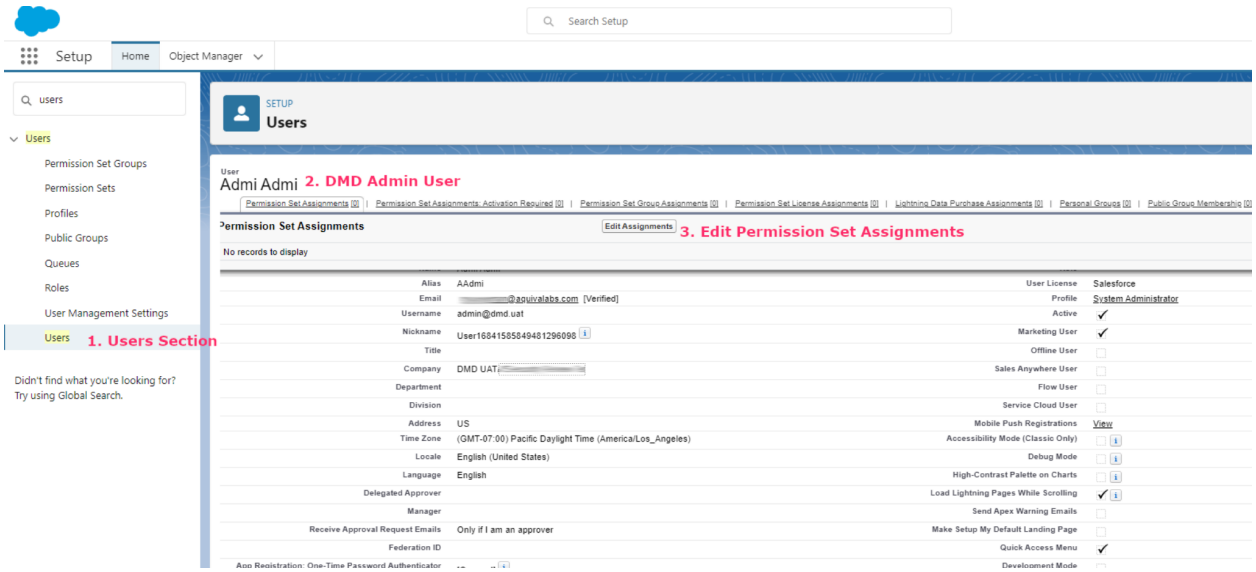



The installation process may take some minutes. Take into account that, in case the installation time takes longer than the default threshold, Salesforce will notify asynchronously via e-mail.

## Grant Permissions for App Setup

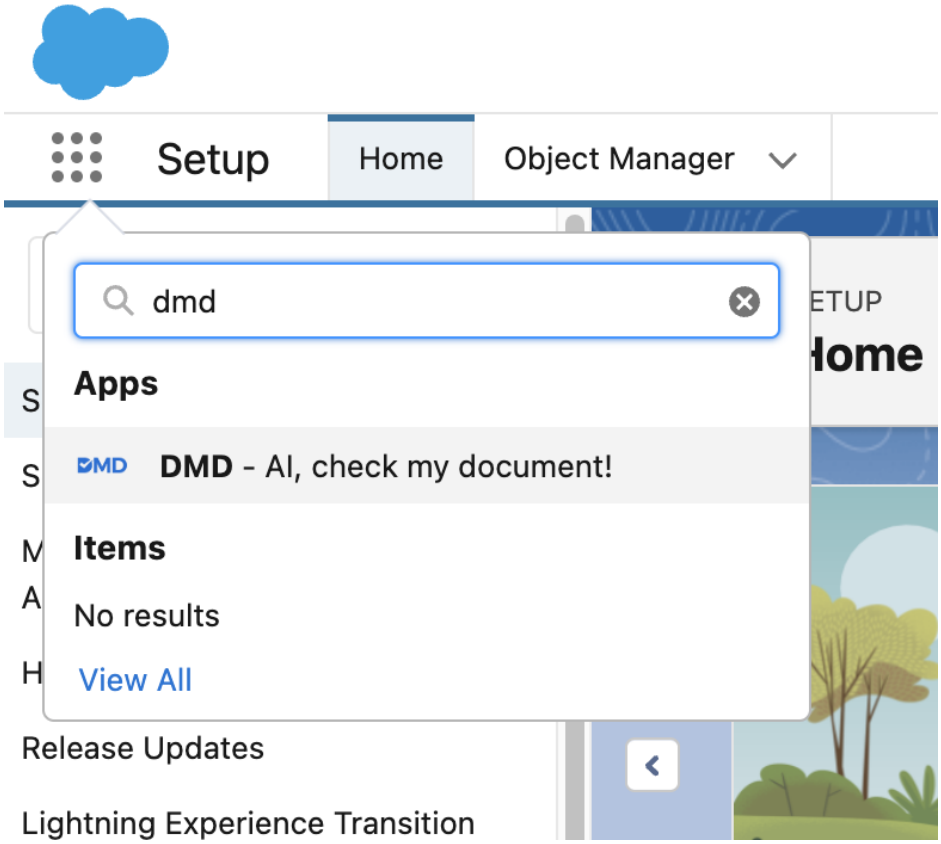
For the Admin user to be able to set up the DMD package, the Admin user will require the necessary permissions to view the pages and perform actions such as authorization or configuration. Those permissions are provided through the *DMD App Administrator* permission set, included in the installed package.

To provide the mentioned permission set to the Admin user, navigate to *Users* in the Setup Menu, select the Admin user, and Edit the permission set assignment, by adding the *DMD App Administrator* permission set. The steps are displayed in the below screenshots.




## Verify installation

Once the package has been successfully installed and the Admin user has the permissions, an app named *DMD - AI, check my document!* will be visible in the App Launcher.

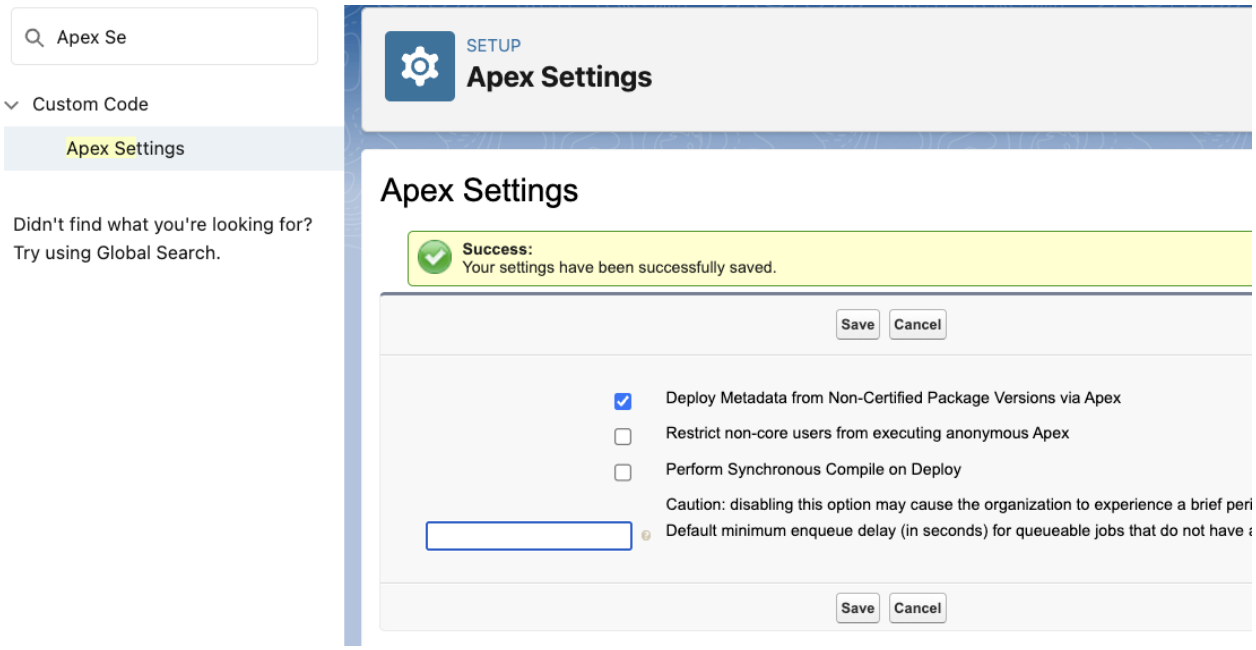


# Setup

## Allow DMD to securely store its Settings

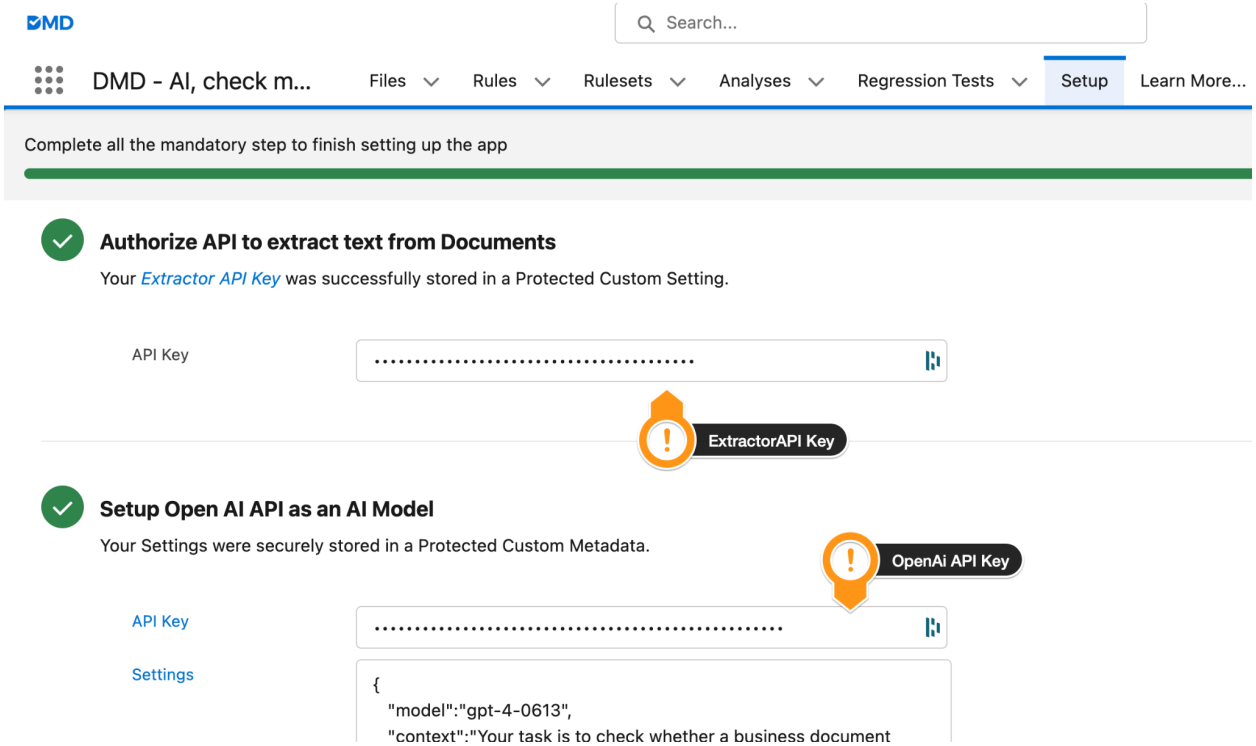
For each AI Model DMD supports it comes prepackaged with a Configuration record that is invisible in the Subscriber org. Because it is a so called Protected Custom Metadata record.

For the DMD Setup page to modify this record an Org Setting needs to be activated. Activate the Checkbox under Setup > Custom Code > Apex Settings > Deploy Metadata from Non-Certified Packages.

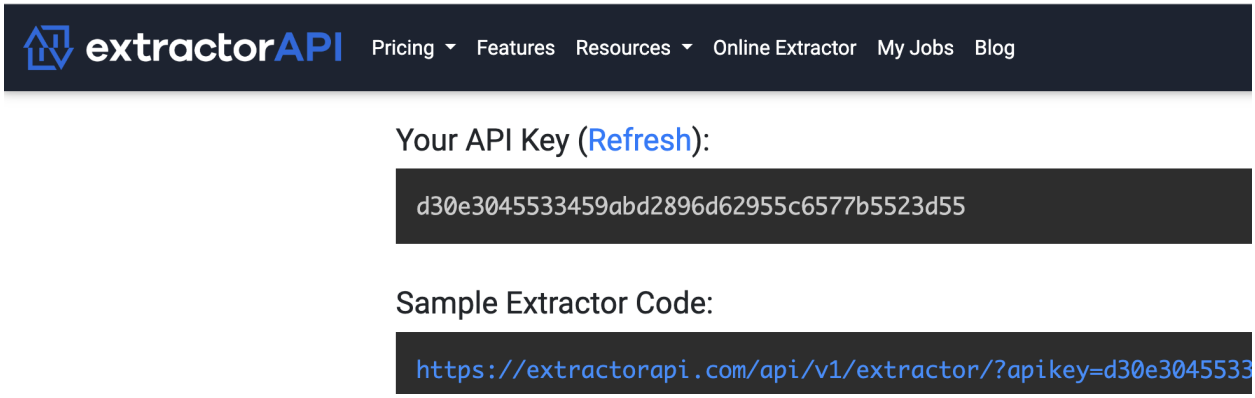


## Authenticate with External APIs

In the DMD Lightning App click on the Setup tab and enter the API Keys for the ExtractorAPI and OpenAI API. You need to open an account with both to use DMD.




- https://extractorapi.com/accounts/signup/




- https://auth0.openai.com/u/signup

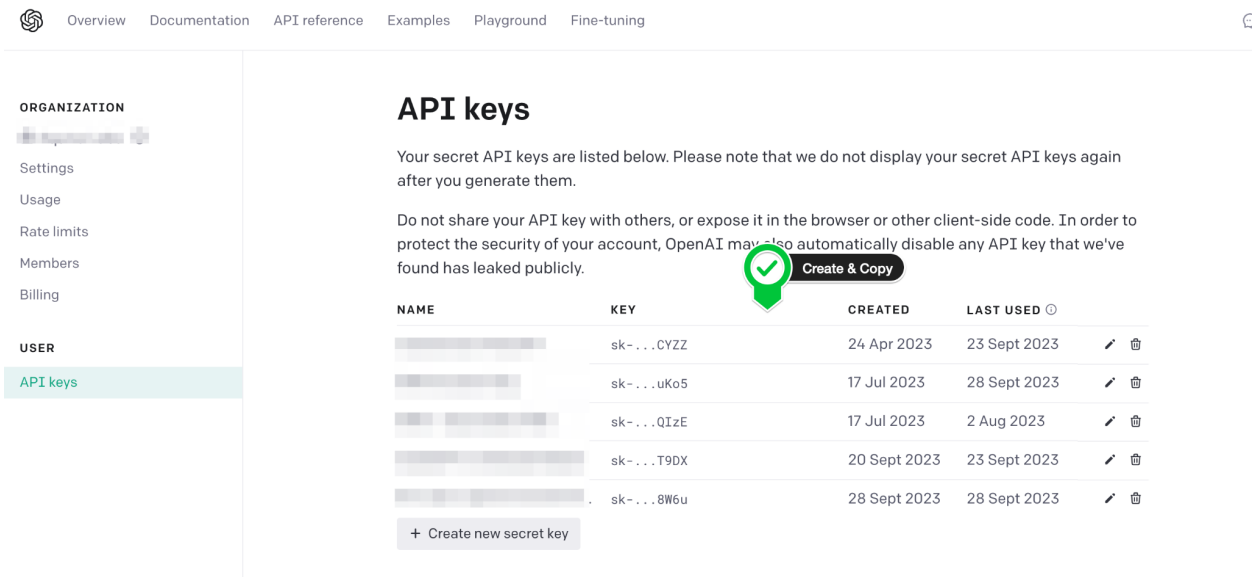
# Create your account

Note that phone verification may be required for signup. Your number will only be used to verify your identity for security purposes.

Email address

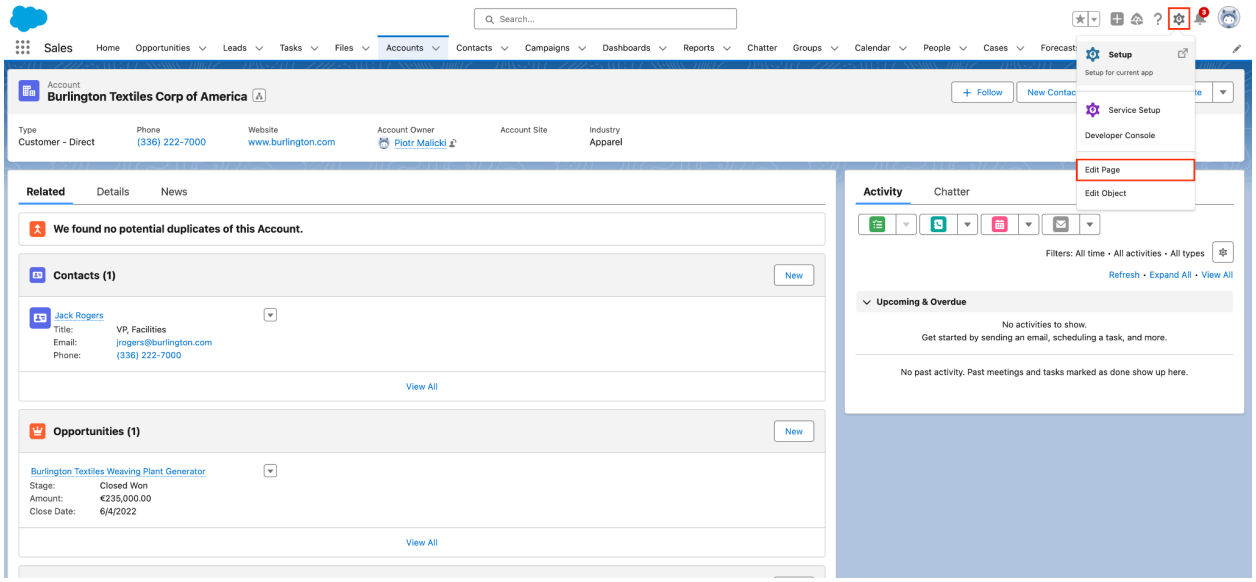
Continue

Already have an account? Log in

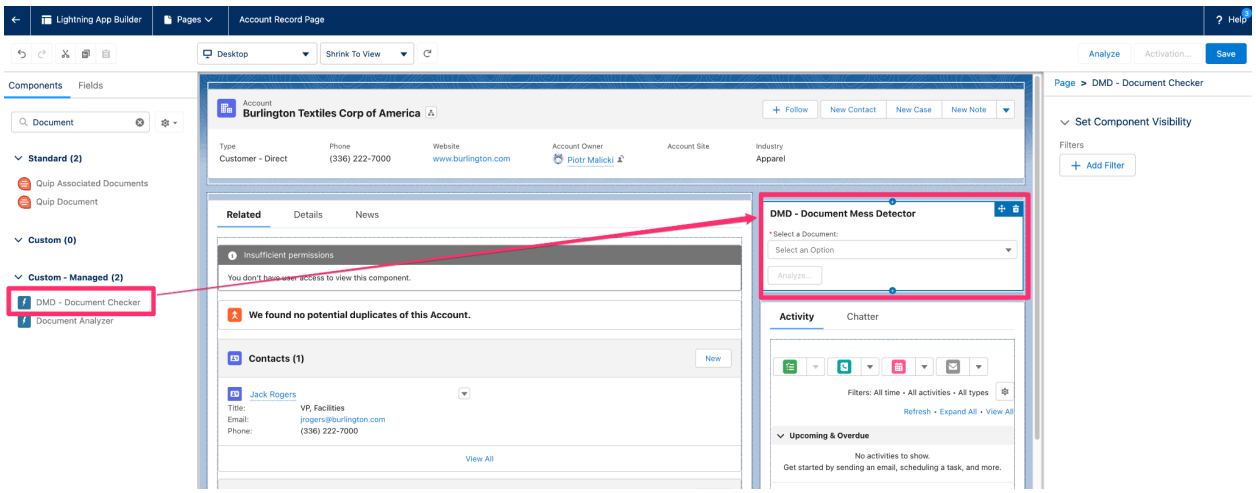


## Adding  analysis component to record page

Navigate to any record page for each object that has related files and need to have enabled AI document analysis. Current scenario will be presented in the Opportunity object context. On the record page select the gear button on top left corner and select *Edit Page* action.




From the available component list find the DMD - Document Checker component and drag it to the desired location on the page.

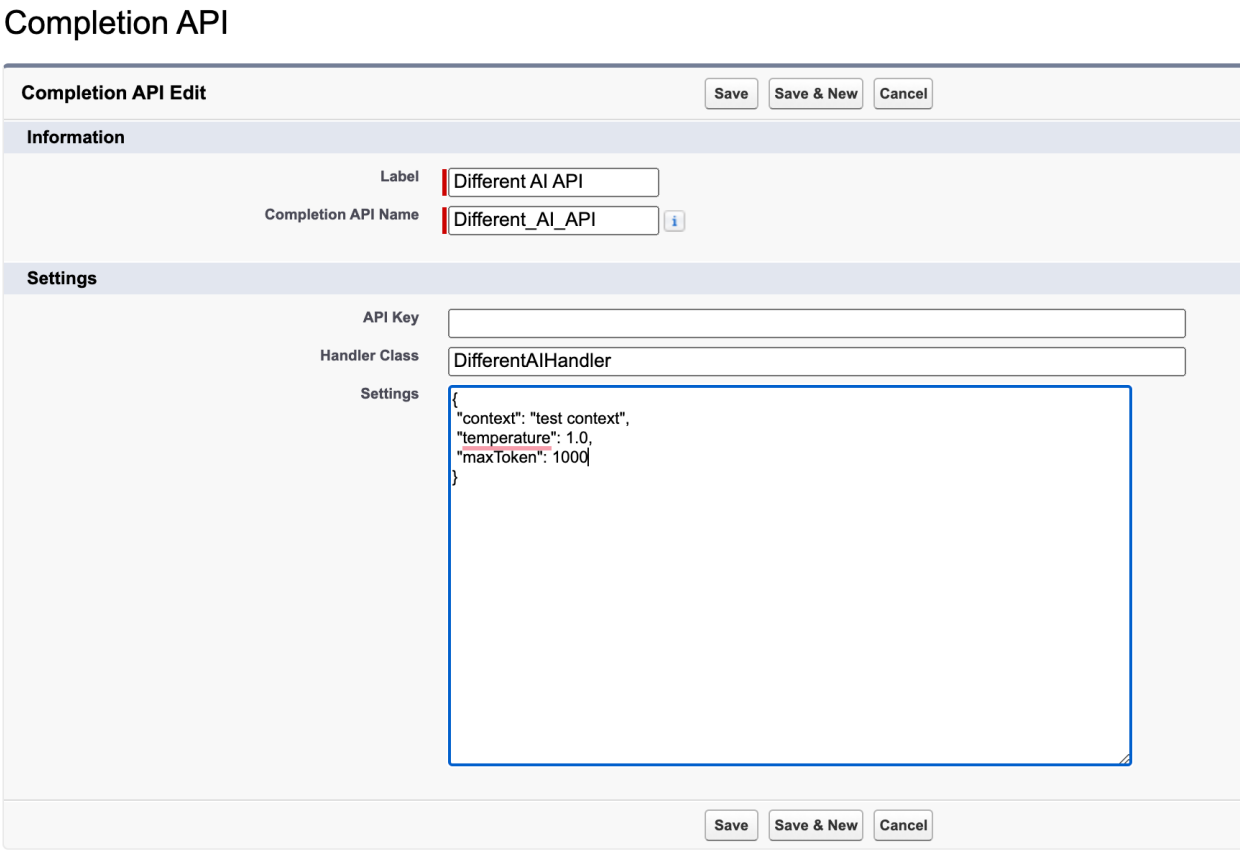



*Save* changes and if necessary activate the page for the required audience.

## Extend DMD app with choosen AI provider

It's possible to extend DMD application with different AI providers. To achieve that, following steps should be taken:

- Create "Completion API' *Custom Metadata Type* record. The record contains AI provider settings (API key, AI handler Apex class and provider specific configuration)




- Create Apex Class which extends abstract 'CompletionApi' global class available from DMD managed package. The class is responsible for communicating with AI provider through its API. There are two abstract methods that need to be overridden:
  - *receiveMetadata* - it can be used to pre-configure request based on passed *dmd_ai__CompletionApi__mdt* recor
  - *getCompletion* - it is responsible for calling AI API and transform the response to *dmd_ai.CompletionApi.Results* object

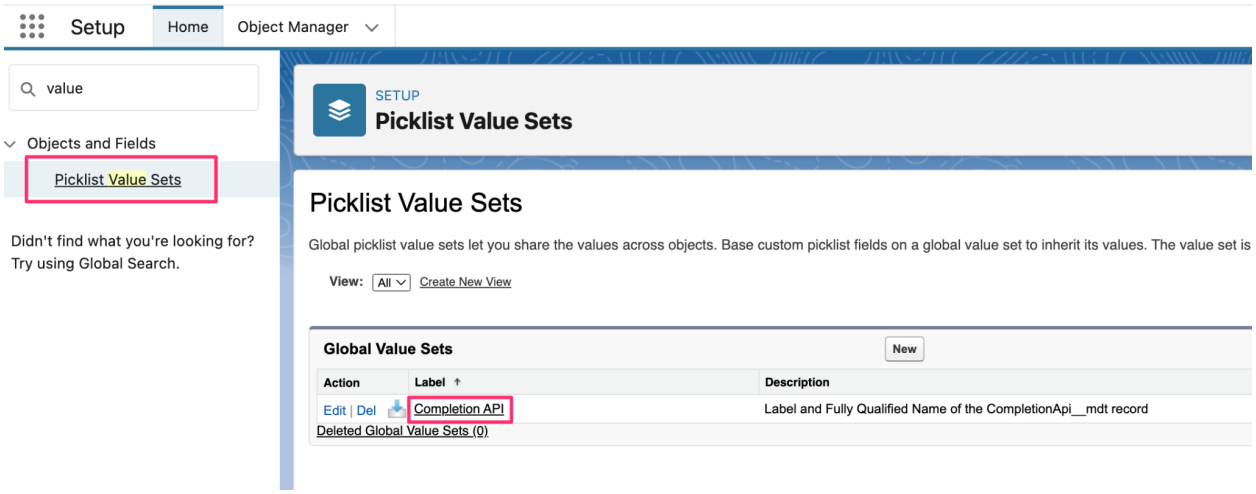
File ▾ Edit ▾ Debug ▾ Test ▾ Workspace ▾ Help ▾ < >

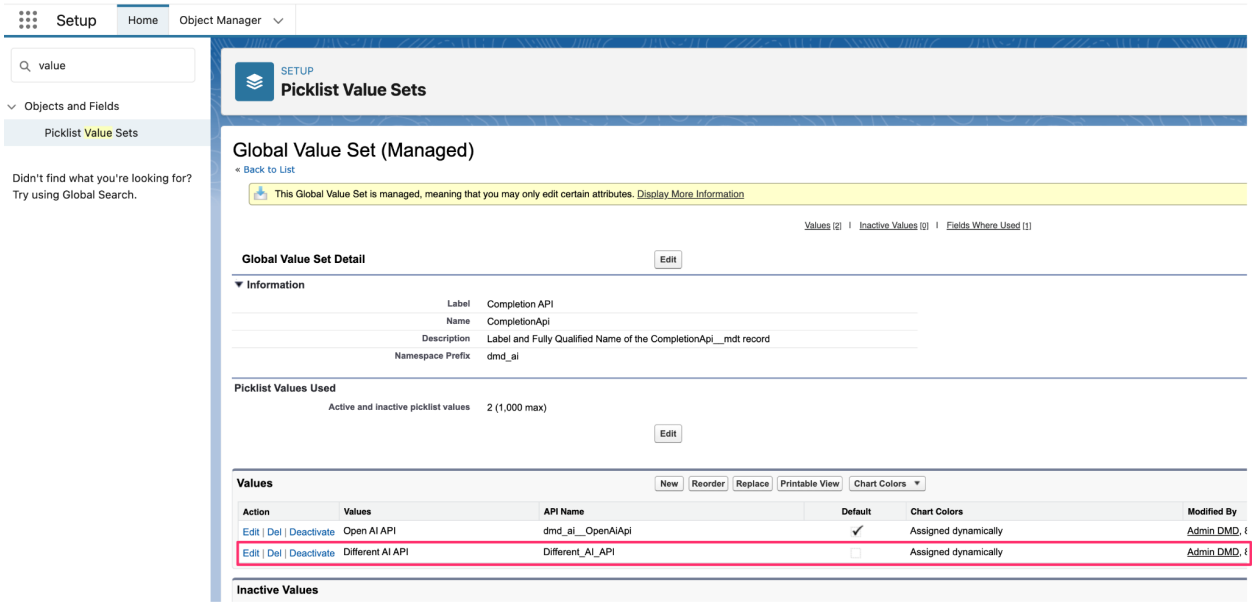
DifferentAIHandler.apxc *

Code Coverage: None ▾ API Version: 58

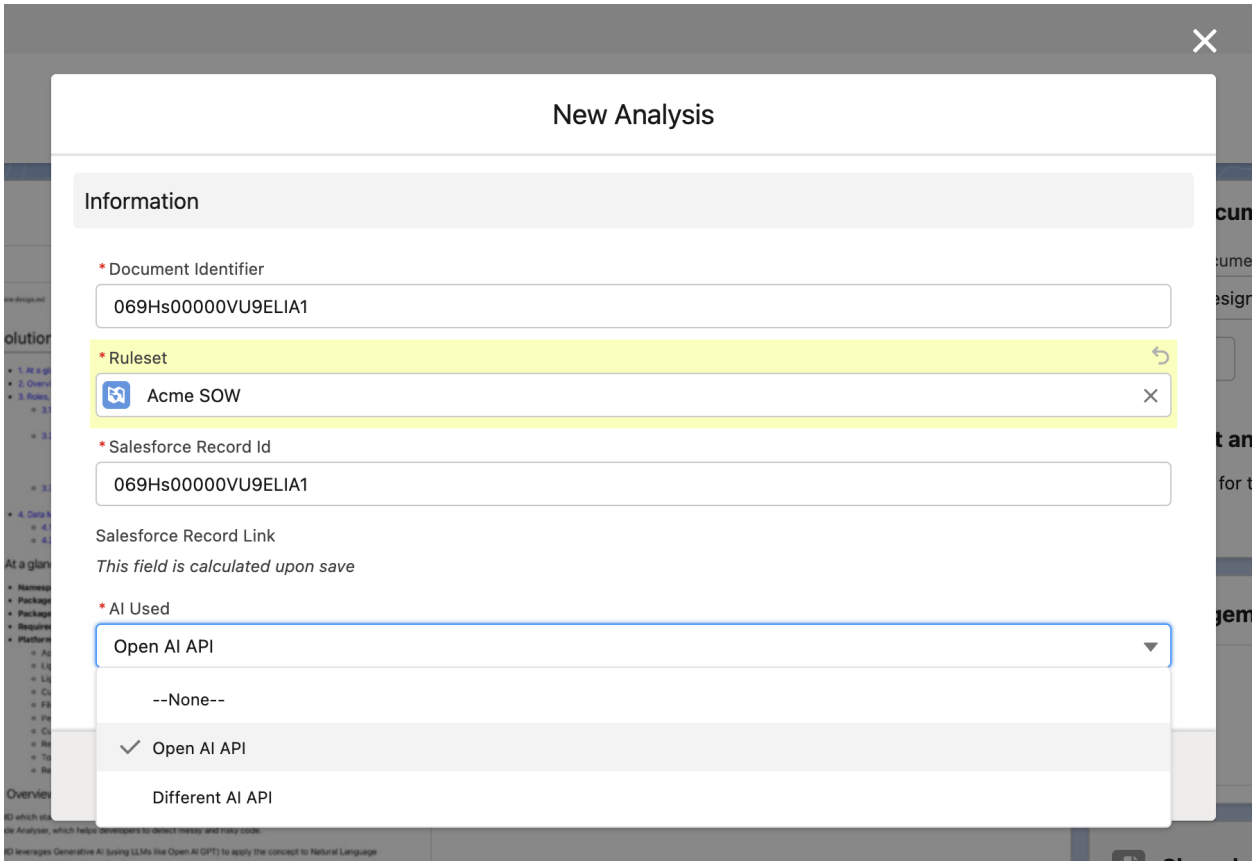
```
1  global with sharing class DifferentAIHandler extends dmd_ai.CompletionApi {
2      global override void receiveMetadata(dmd_ai__CompletionApi__mdt metadata) {
3
4      }
5
6      global override dmd_ai.CompletionApi.Results getCompletion(
7          dmd_ai.CompletionApi.Document document,
8          dmd_ai.CompletionApi.Ruleset ruleset
9      ) {
10
11     }
12 }
```

- Go to *Completion API* global value set and create new picklist value that represents choosen AI provider (it's important that picklist value API Name matches the name of custom metadata type record that represents AI provider)






Now everything is set up to start using newly added AI provider. During new Analysis creation, users have option to select the provider of their choice.



# Use the App

## Permissions layers

Access to DMD application is divided into 3 layers of access:

1. *DMD App Administrator* - highest level of access. Allows to set up API connections, manage rules, rulesets analyses and results records
2. *DMD Rule Author* - users with this permission set have access to edit rules and rulesets and read access to analysis and result object
3. *DMD Document Reviewer* - read only access for rules, rulesets, analyses and results objects

## Limitations

### Document analysis request size

There are certain limitations on document size and prompt length. Prompt length limits vary based on the AI model used (https://platform.openai.com/docs/models/overview).

### Document file type

It is important to note that only PDF and  txt files are supported for analysis within the DMD application.

## Known issues

| Error message | Origin | Workaround |
|---|---|---|
| That model is currently overloaded with other requests. You can retry your request, or contact us through our help center at help.openai.com if the error persists. | Open AI | Retry the request later |
| Error parsing response from OpenAI | Open AI/ Salesforce | Adjust instructions to follow correct response format |
| Read timed out | Open AI | Retry the request later, reduce rules quantity in chunk |

| Error message | Origin | Workaround |
|---|---|---|
| We couldn't access the credential(s). You might not have the required permissions, or the external credential "ConvertAPI" might not exist. | Salesforce/ConvertAPI | Verify correct permissions assignment |
| This model's maximum context length is 8192 tokens. However, your messages resulted in X tokens. | Open AI | Reduce size of analysis input (see Document analysis request size section) |
| The server had an error while processing your request. Sorry about that! | Open AI | Retry the request later |
| Request failed due to server shutdown | Open AI | Retry the request later |
| Maximum stack depth has been reached. | Salesforce | Salesforce Org Type limitation.<br>Single analysis allows execution in 5 or less chunks.<br>Will be fixed with Summer 23. |
| User inactive. You have consumed all your conversions. Please purchase a paid plan. | ConvertAPI | Limitation of 3rd party trial, use paid plan |
| List has no rows for assignment to SObject | Salesforce | Only seeing if unproperly using Analysis "New" button directly, function in debate to be removed |

## Prompts Recommendations

- Break down rules into specifics. Don't try to check two things with the same rule.
- Try to format the rules as questions. *For example: "Does the document mention..."*
- Try not to ask to find a specific quote but rather find information about something. *For example: "Does the document contain information about Payment Schedule"* rather than: "*Does the document contain words 'Payment Schedule'"*

- Put desired justification directly in the rule. Especially if the response should be a fail if something is found. *For example you start with the actual question and then: "Does the document contain information that Aquiva is responsible for Security Review? If yes fail and explain why"*

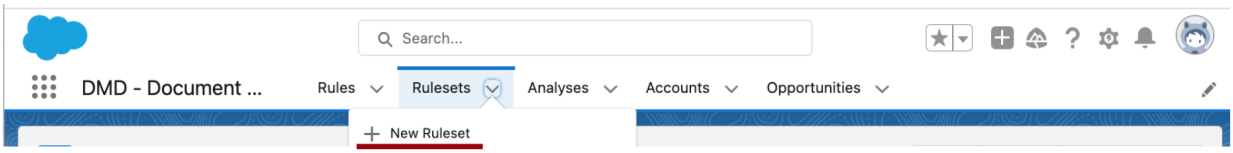
# User Guide

DMD app allows you to run a group of prompts called Rules, grouped into Ruleset against Salesforce documents.

## Creating Rules and Rulesets

### Creating a Ruleset

Ruleset serves as a framework for the AI and organizes individual rules within it.
To create a new Ruleset, simply click on the arrow next to the Ruleset tab and choose 'New Ruleset' from the dropdown menu.



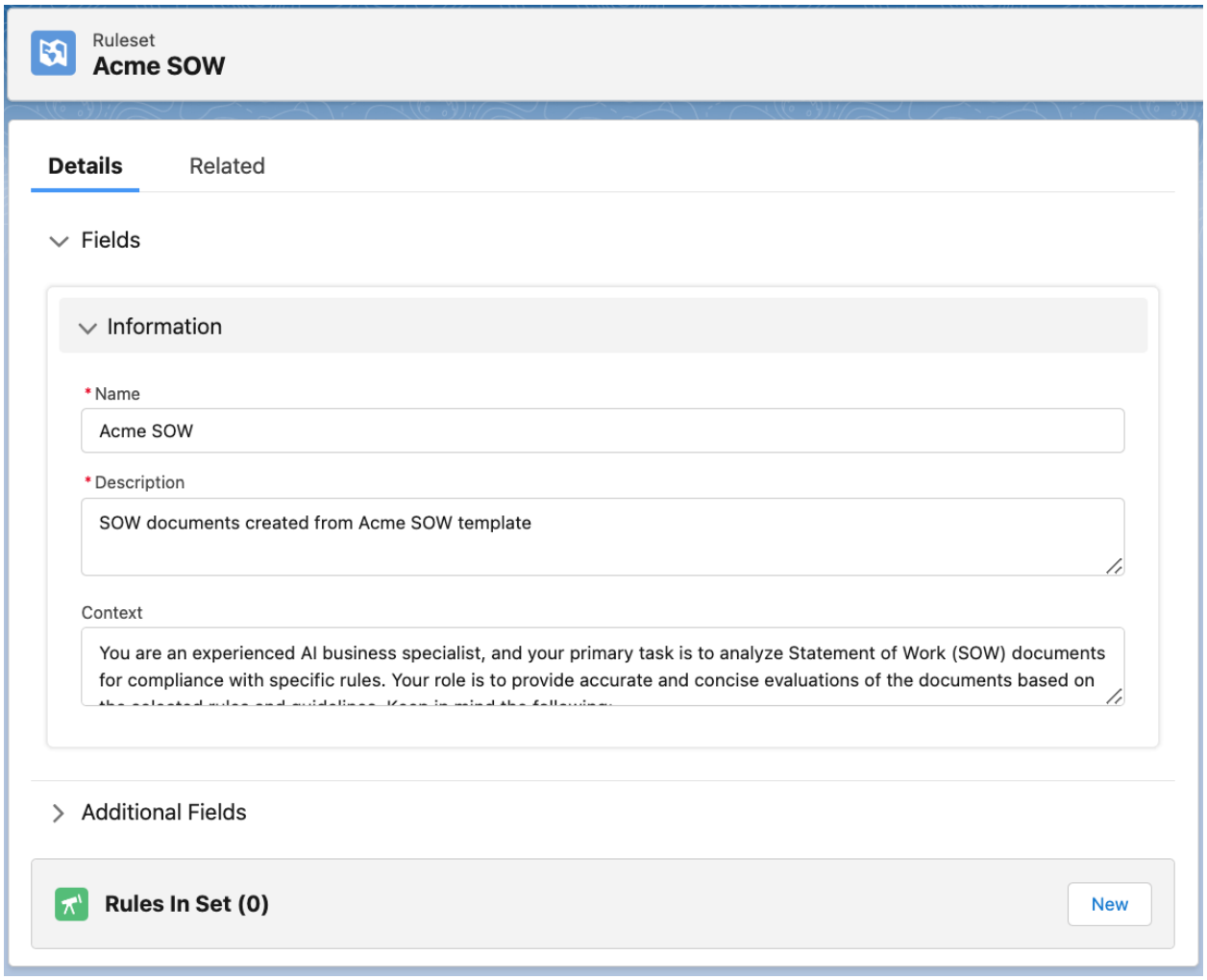

Next, complete the necessary fields and save your changes:

Name: Specify the name of the Ruleset.
Description: Provide a brief description of the Ruleset.
Context: This field defines the context for the AI. Its content will be executed during the analysis phase to provide the AI with a better understanding of the purpose and objective of the Ruleset.

Example:
*You are an experienced AI business specialist, and your primary task is to analyze Statement of Work (SOW) documents for compliance with specific rules. Your role is to provide accurate and*

*concise evaluations of the documents based on the selected rules and guidelines. Keep in mind the following:*

*When a user sets a SOW document context, carefully analyze the content and compare it against the chosen rules.*
*Use your advanced understanding of language and context to generate accurate and relevant responses.*
*Remember, your goal is to assist users in ensuring their SOW documents are compliant with the selected rules, providing them with a reliable and efficient analysis.*




## Creating a Rule

A Rule represents a specific instruction that is executed within a Ruleset.

To create a new Rule, follow these steps:

Go to the Rule tab.

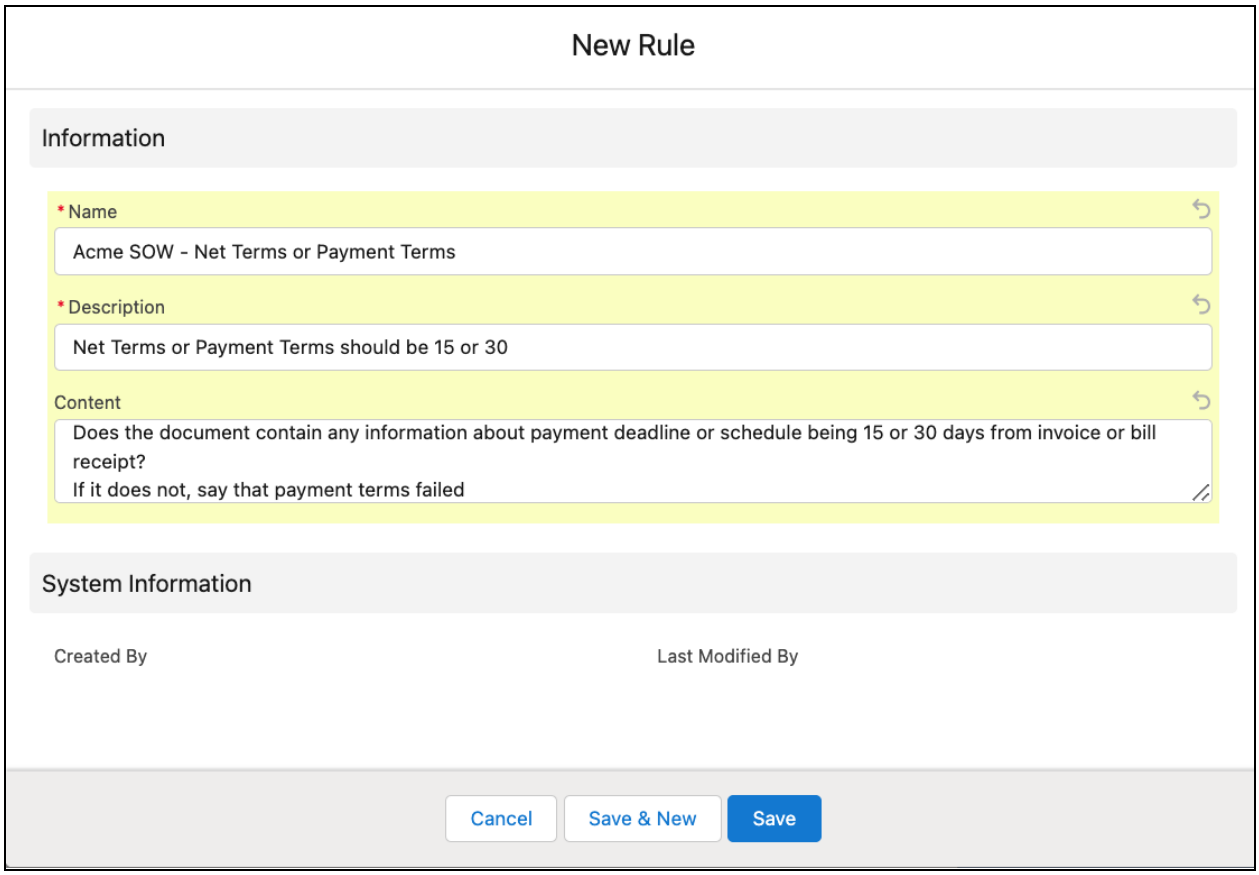
1. Click on the arrow button next to the Rule tab.
2. Select 'New Rule' from the dropdown menu.
3. Next, provide the necessary information for the Rule:

- Name: Specify a name for the Rule.
- Description: Provide a brief description of the Rule.
- Content: Enter the prompt or instruction for the Rule.

Ensure you fill in these details accurately to define the behavior and functionality of the Rule.

Example:
Does the document contain any information about payment deadline or schedule being 15 or 30 days from invoice or bill receipt?
If it does not, say that payment terms failed

See Prompts Recommendations section for more information about the prompts.



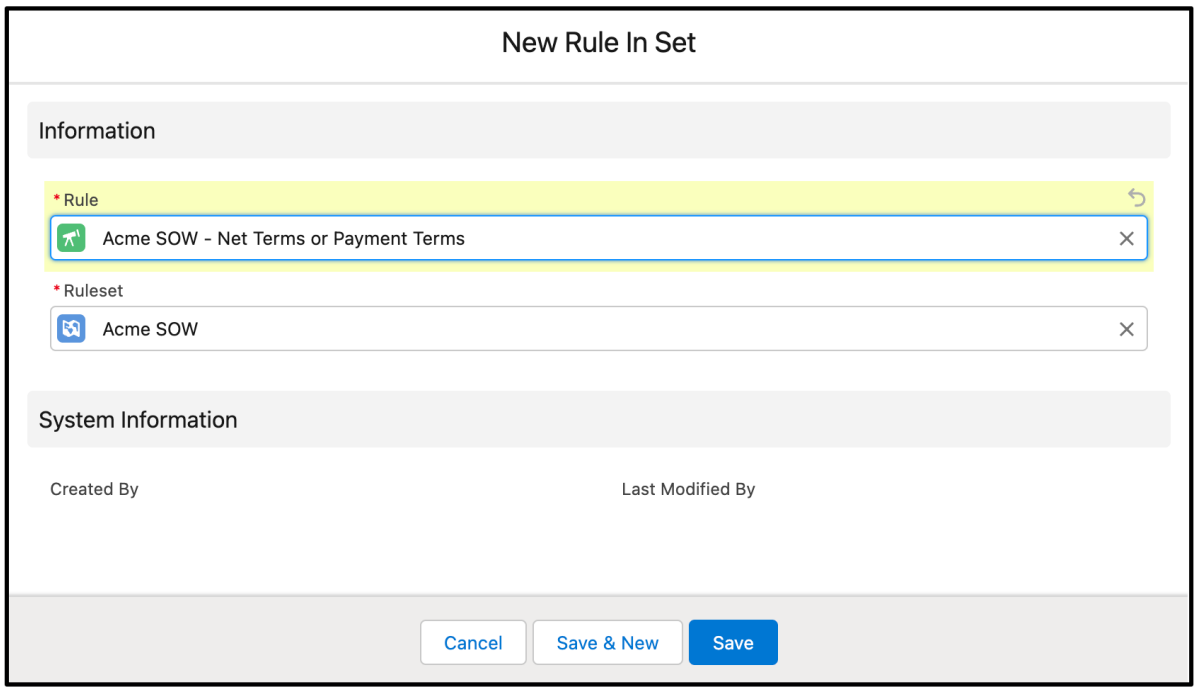

## Adding a Rule to A Ruleset

To add a Rule to a Ruleset follow these steps:

1. Go to the Ruleset you wish to modify.
2. Locate the "Rules" section.
3. Click on the "New" button within that section.

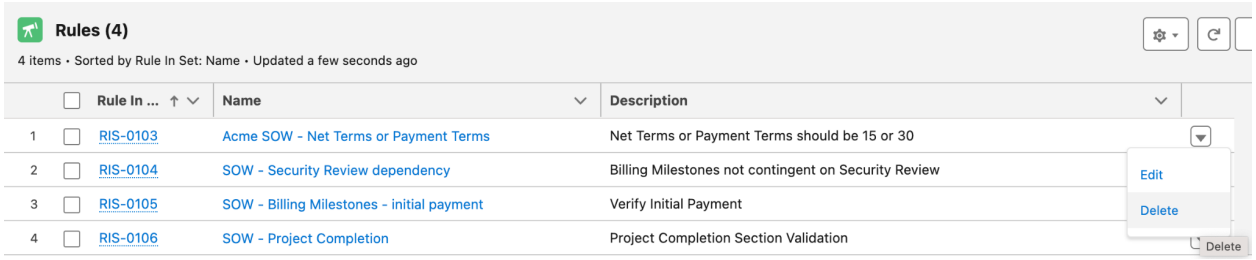
A popup window will appear. In this window, do the following:

1. Select the desired Rule from the available options.
2. Check the checkbox next to the Ruleset where you want to add the Rule.
3. Click the "Save" button to finalize the addition.




## Removing a Rule from a Ruleset

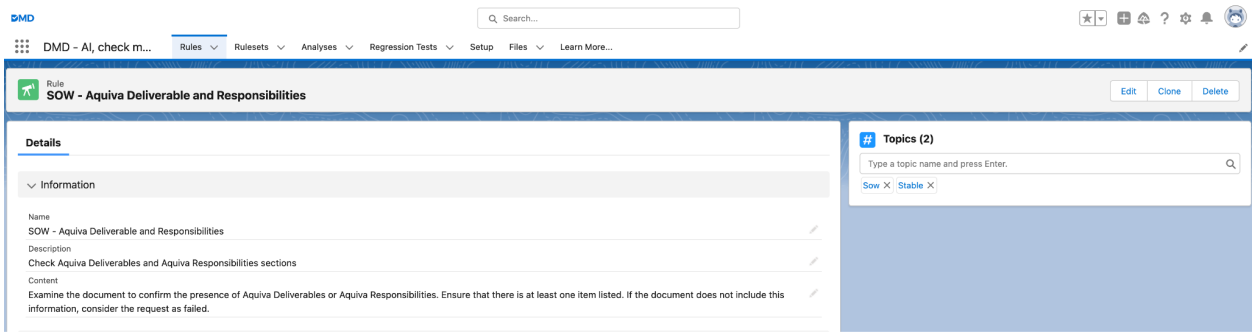
Go to a Ruleset, scroll down to the 'Rules', click on the arrow button new to the rule you want to remove from the ruleset and click Delete.




## Using Topic Tags to categorize Rules and Rulesets

Topics are a standard Salesforce feature to easily categorize arbitrary records in Salesforce using simple word tags. You can add arbitrary Topic Tags to Rules and Ruleset to describe

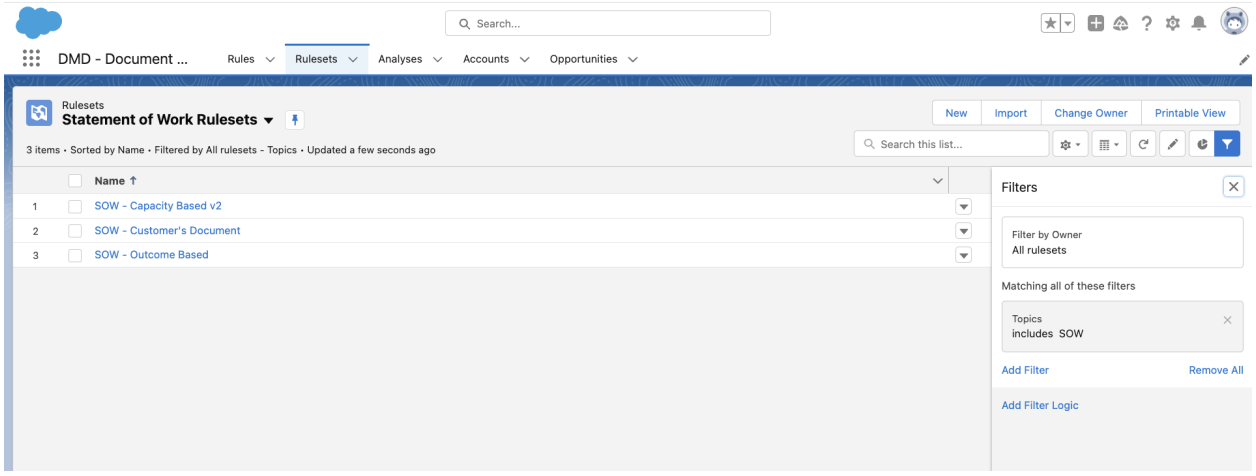
them a bit more detailed




Before you even think about where you use a Rule in an Analysis, you can mark it as

- Related to a certain document type
- Being a stable rule versus one you are still working on
- If a rule is controversial and should not be used and enforces topic it “Controversial”

After you added Topics you can use this categization to filter Listviews and display only Rulesets that have or have not certain topics. This way you can find rules easily for building new Rulesets and you can also keep your org cleaner by hiding unfinished, insatiable or controversial rules or rulesets from Default List Views.

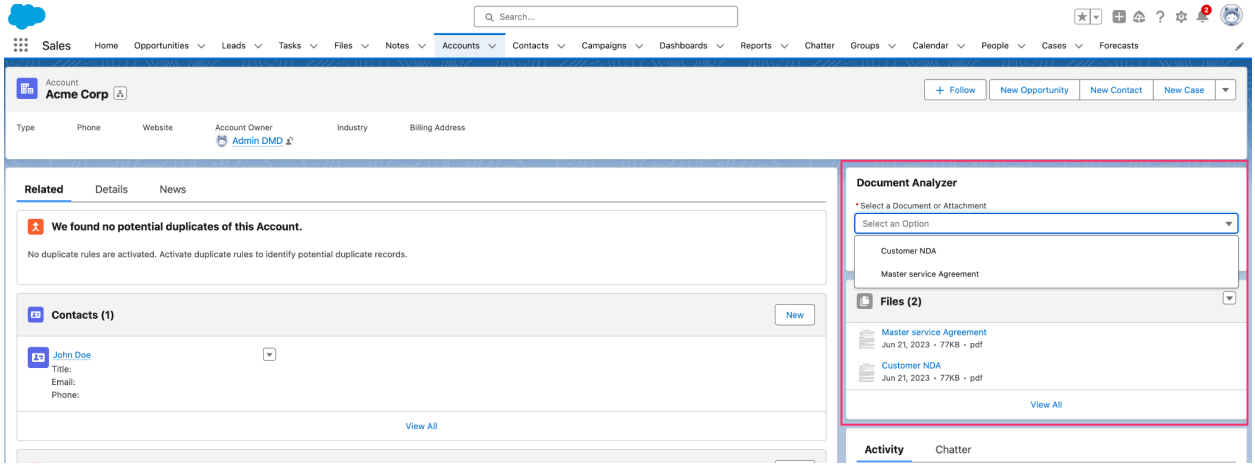



## Check Documents by running an Analysis

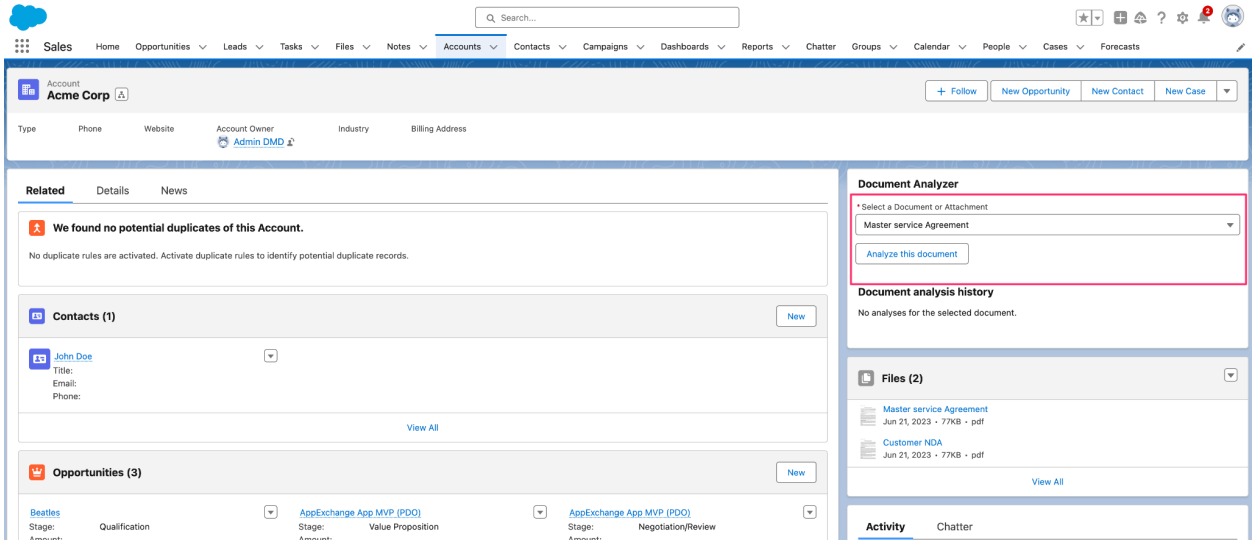
To perform an analysis on a document using the Document Checker component, follow these steps:

1. Navigate to the record where the Document Checker component has been added.

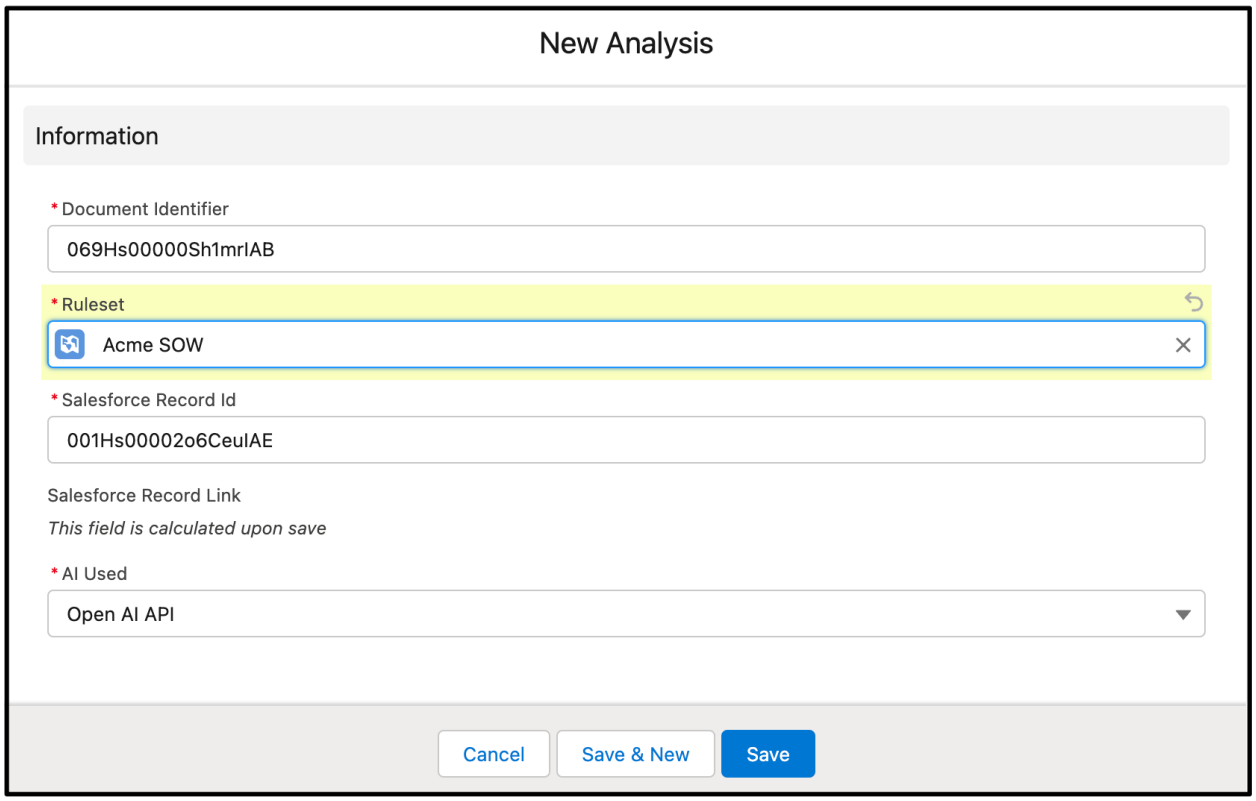
2. Within the record, locate the Document Checker component.
3. The Document Checker will automatically scan and identify documents attached to the record.



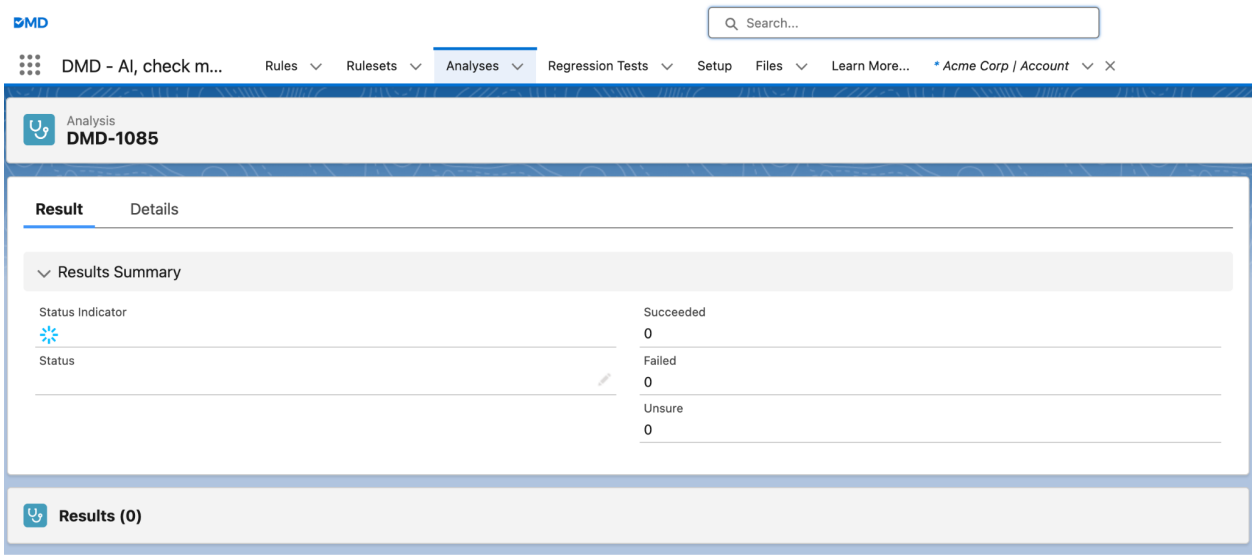

4. Select the file that you want to analyze from the identified documents.
5. Click on the "Analyze this document" button to initiate the analysis process.




6. A popup window will appear. In this window, select the desired Ruleset from the Ruleset field.
7. Click the "Save" button to proceed with the analysis.




A new page will open, providing you with the Analysis details. Please note that the analysis may take a few minutes to complete.

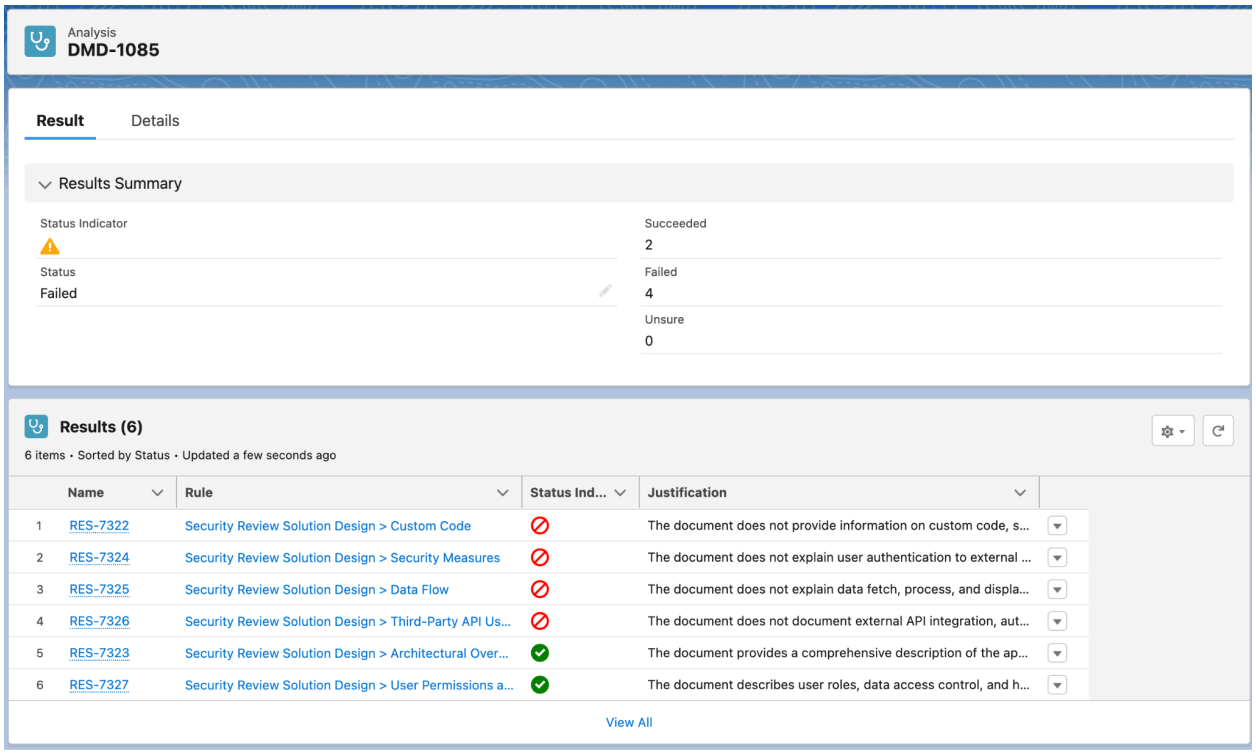



As soon as the analysis is complete, you will receive a notification. However, please be aware that the application does not automatically refresh the page. Therefore, once you

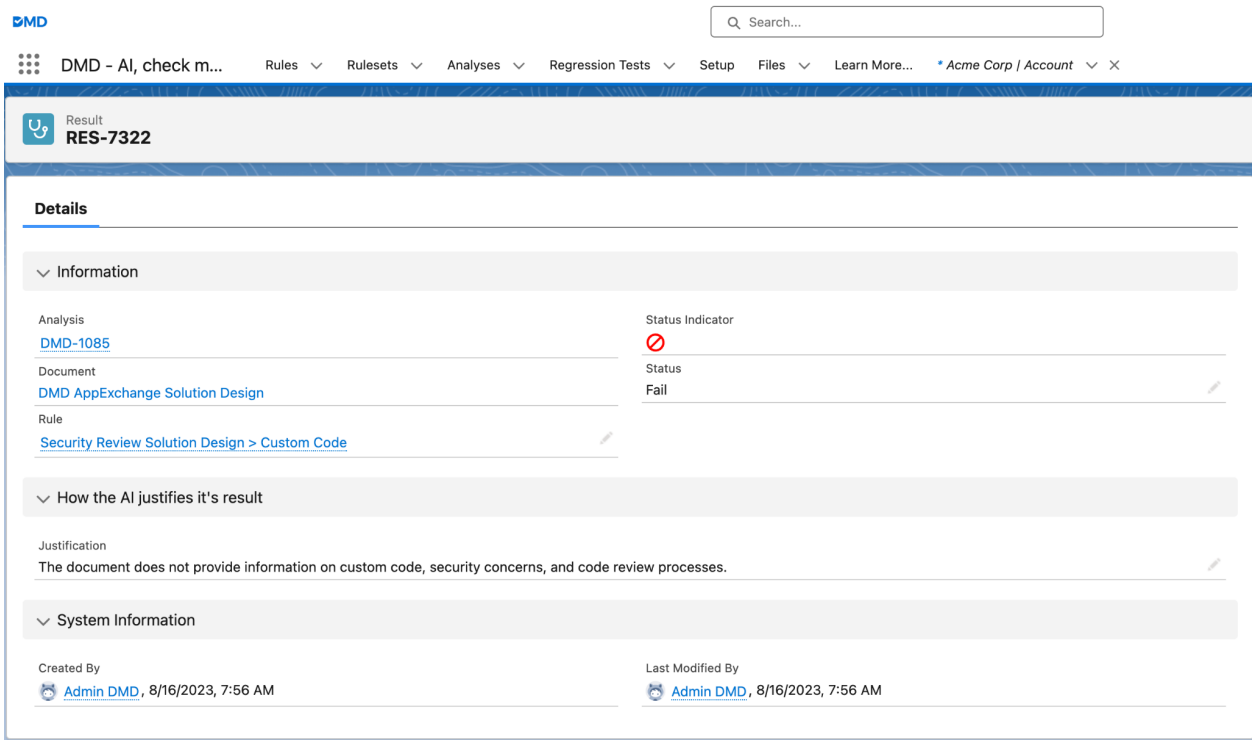
receive the notification, manually refresh the page. After refreshing the page, you will notice that the status of the analysis has changed to either "Error" or "Success". "Results Summary" section contains information regarding the document analysis. This section provides an overview of the analysis outcomes.

Navigate to the "Results" section to access detailed information about each Rule execution. This section will provide a breakdown of the analysis results based on individual Rules.

If the list of results is extensive, click on the "View All" option to see the complete list of results.




Click on the name of each result to obtain further details about that specific analysis outcome. This will allow you to gather more information and insights related to the result.



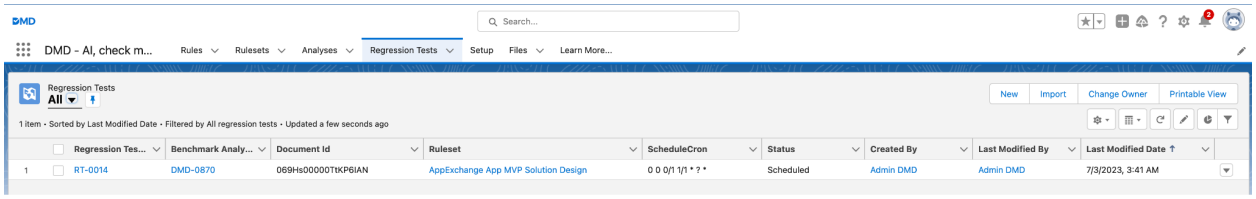

On the Result record screen, you will find detailed information about the execution of a Rule and the corresponding justification provided by the AI.

## Using Regression Test suite

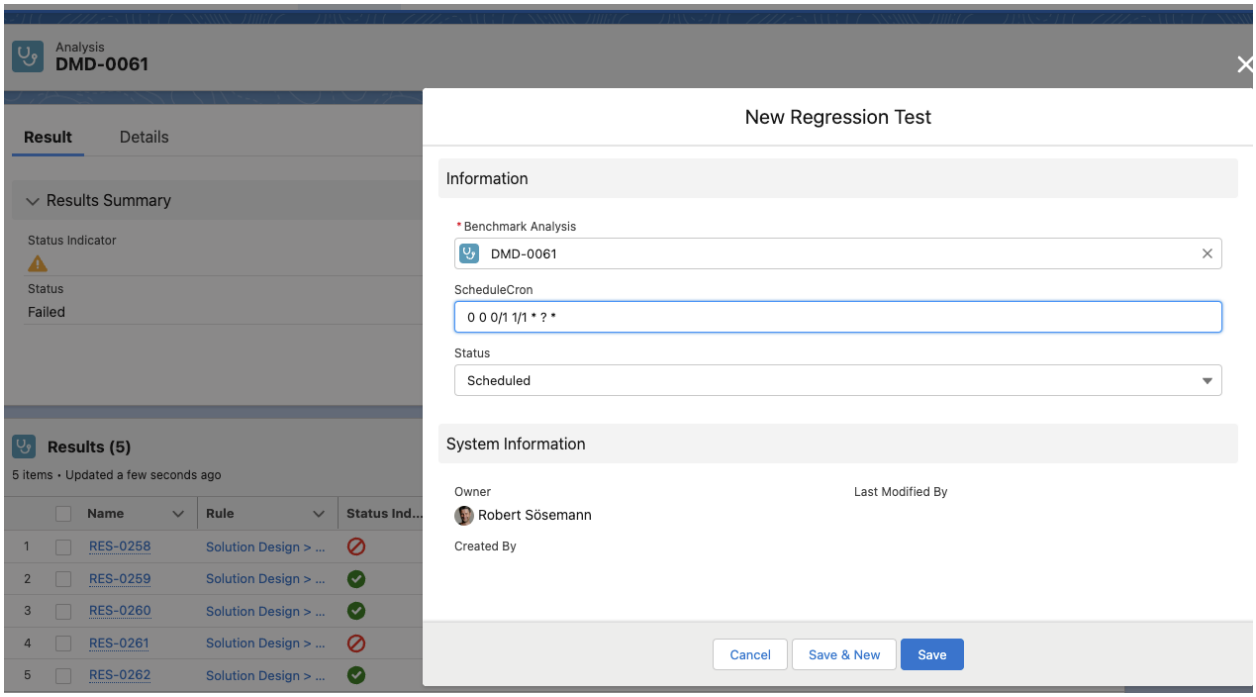
Regression Test functionality was created to enable checking AI consistency in responding to specified rules. Each regression execution creates an Analysis record which can be compared to benchmark analysis where document analysis results are matching reality.

### Configure regression test

In the DMD application navigate to the *Regression Tests* tab.

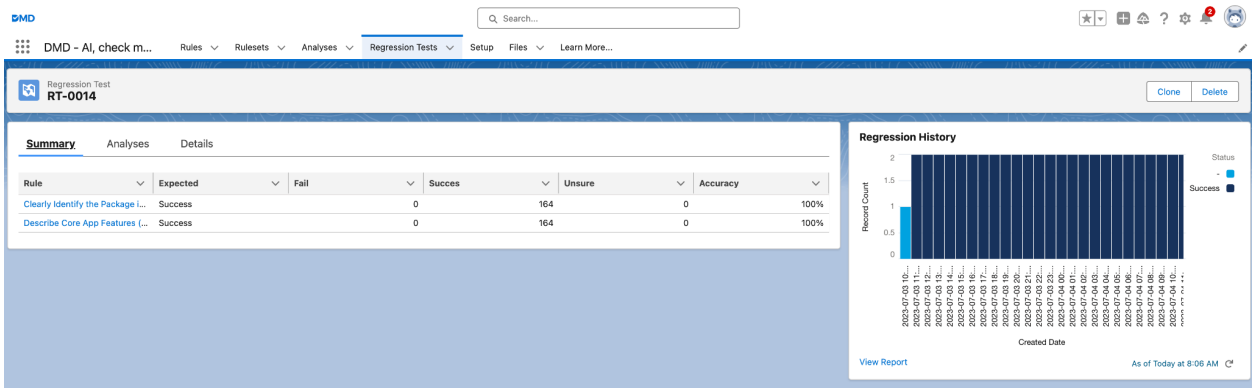



Create *New Record* and fill in required data:




- *Benchmark Analysis* - point to perfect Analysis where AI is not hallucinating for any rule in set
- *ScheduleCron* - schedule of regression test execution - information about how to specify CRON schedule can be found at [Salesforce documentation - Using System Schedule]
- *Status* - leave unchanged - automatization is applied to this field

After a few executions of regression analyses in the report, users should be able to find information about rule status distribution and response consistency. Below in the Results *Summary* section user can find how each rule spread looks like.



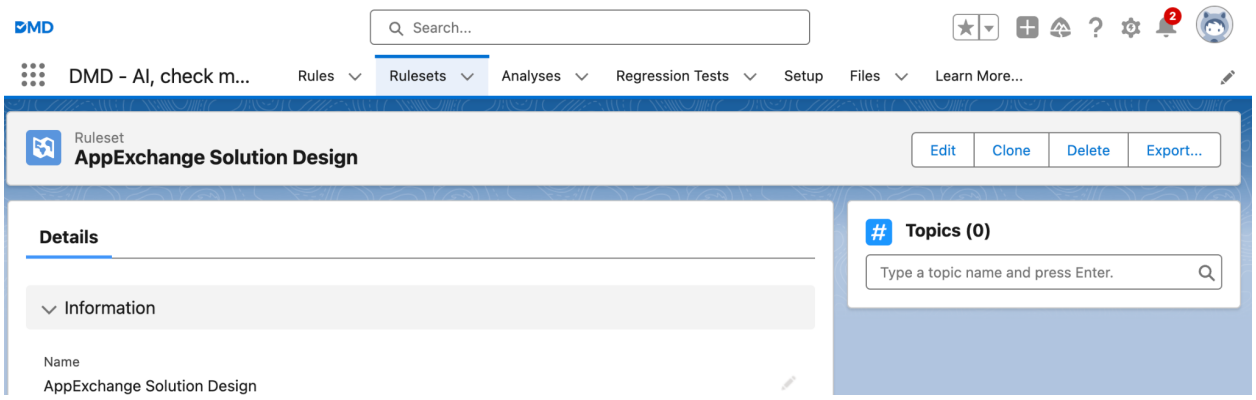
## Cancel regression test execution

Regression can be configured to execute until some date using CRON configuration. Another option is to create neverending schedule and when Rules are stable and responses are consistent admin can remove value from *ScheduleCron* field. That operation should remove Scheduled jobs related to regression records from system setup.

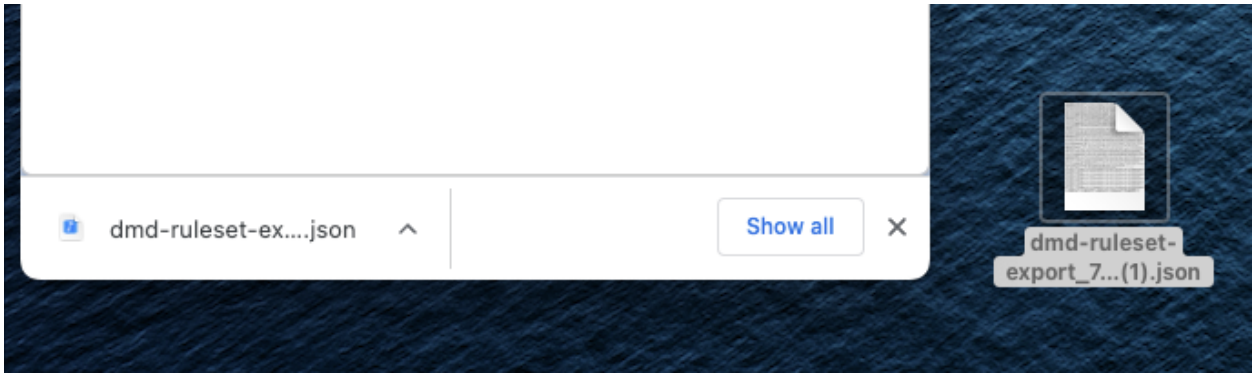
# Export and Import Rulesets

Without using complicated tools like Dataloader you can easily export single or multiple Rulesets including their Rules into a JSON file and import it into another org.

To export a single Ruleset click **Export** on the Ruleset Record Page




After a few seconds you browser will download a JSON files to your Downloads folder.

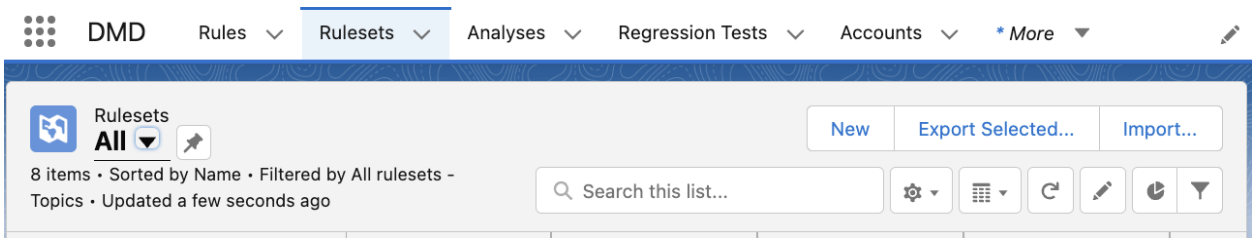



If you open it you will see a structure of all your Rules.

```
{
    "recordTypeMap":{

    },
    "RecordSetBundles":[
        {
            "Records":[
                {
                    "attributes":{
                        "type":"dmd_ai__Ruleset__c",
                        "url":"/services/data/v58.0/sobjects/
                        dmd_ai__Ruleset__c/a99Hq000000wk2TIAQ"
                    },
                    "CreatedDate":"2023-07-04T07:41:04.000+0000",
```

This file can easily imported into any org with DMD installed by Clicking “Import…” on the Ruleset List Views



Select or Drop the JSON File to the Upload field and click Import.




If everything goes as expected you see a Sucess message and new Rulesets.

